5.5. Restaurant

Description

The robot is required to retrieve and serve orders to customers in a real, previously unknown restaurant environment.

Main goal: Detect calling or waving customer, autonomously navigate to their tables, take their orders, and serve them accordingly.

Optional goal: Use an unattached tray to transport the order.

Focus

This task focuses on *Task planning*, *Online mapping*, *Navigation in unknown environments*, *Gesture detection*, *Verbal interaction* and *Object manipulation*.

Setup

- **Locations:**
  - This task takes place in a real restaurant fully equipped and in business. If this is not possible, an alternative venue may be selected, provided it resembles a restaurant environment and is outside the *Arena*.
  - The *Restaurant* location will remain undisclosed until the start of the test.
  - The robot starts next to the *Kitchen-bar* (a designated table located near the restaurant's kitchen), facing the customer seating area (dining tables).
- **People:**
  - A *Professional Barman* (member of the *technical committee* (TC)) awaits at the other side of the *Kitchen-bar* for orders to be placed. The *Professional Barman* assists the robot on request.
  - There may be real customers and staff present.
  - There are at least three tables occupied by professional customers (member of the *organizing committee* (OC), *technical committee* (TC) or other teams).
  - There are at least two tables occupied with regular customers.
  - Customers may call the robot any time, even simultaneously.
- **Furniture:** The furniture is not standardized and remains unchanged during the test.
- **Objects:**
  - Objects to fulfill orders are located on the *Kitchen-bar*.
  - Each order consists of 2–3 objects, randomly selected.
  - All edible/drinkable objects from the list of standard objects (see Section 3.2.5) are eligible to be part of the orders.

Procedure

1. The referee instructs the team to bring the robot to the start location.
2. The referee gives the start signal and starts the timer.
3. The team leaves the area after the start signal.

4. A *technical committee* (TC) member follows the robot ready to press the emergency stop button.
5. The robot detects a calling or waving customer and navigates to their table.
6. The robot take the customer's order, place the order, and deliver it.
7. **Optionally**, the robot may use an unattached tray to transport the order.

Additional rules and remarks

- **Remarks:**
  - This test occurs in a public area. Any physical contact with people or furniture results in immediate emergency stop.
  - Since this task is performed outside the arena, the time limit may be longer than the others tasks.
  - The availability of wireless, external computing devices, or electrical outlets can't be guaranteed. Assume unavailability.
  - The robot must interact with the customers directly. Teams are not allowed to assist or instruct them.
  - The robot may use up to two minutes to instruct the *Professional Barman* per order.
  - Examples of instructions include, but are not limited to:
    - Request guidance to a customer's table.
    - Ask for pointing direction.
    - Ask to press on the screen.
  - The robot may:
    - Take and place several orders before delivery,
    - Alternate between order-taking and serving, or
    - Handle one order at a time.
  - The robot should politely confirm the order to the client when receiving it, keeping the guest pleased.
  - The robot can either transport each object individually, or using a tray. All delivered objects must be placed on the customer's table.
  - For tray delivery, the robot must:
    - Place the items on the tray,
    - Pick up and transport the tray,
    - Deliver the items by either directly placing them on the table or placing the tray first.
  - If requested, the *Professional Barman* will place the order in a basket or tray for the robot to deliver it.
  - Only two team members are allowed near the robot upon arrival for watching and charging.
  - If audience interference makes task execution impossible, the team may immediately repeat the test.
  - Each Deus Ex Machina penalty for skipping manipulation is capped at twice per order so receiving an order with three objects is not more punishing.
  - If the robot detects a customer but does not reach their table, it must clearly identify the person (e.g., show a picture) to earn partial points.

  - When at the front of the queue, teams may begin startup procedures with the robot stationary. Once called, the robot must be brought directly and steadily to the start location—only minor adjustments are allowed (no back-and-forth or full turns).
- **Disqualification:**
  - Touching the robot after the start signal.
  - Mapping the environment in advance.

Instructions:

To Referee

The referee needs to:

- Prepare orders for each client.

To OC

The OC needs to:

- **During Setup days:** Check with local (security) management if the possible location, including a sufficient queuing area, can be used for the restaurant test.
- **1 hour before the test:** Gather all teams and robots to move to some nearby queuing area and instruct the teams how/when to move to the actual test location.